The natural domain of $f(x)$ is $[1, \infty[$. Not any point in this interval, however, belongs to the domain of the composite function $h(x)$. Since the expression $\sqrt{y-2}$ is meaningful only if $y \geq 2$, and for $y=2$ we have $x=5$, the natural domain of this composite function is represented by $[5, \infty[$, i.e. a subset smaller than the natural domain of $f(x)$.

Let us examine one more example of a composite function. Consider the function $y = \sin(\arcsin x)$. You know that $\arcsin x$ can be regarded as an angle the sine of which is equal to x. In other words, $\sin(\arcsin x) = x$. Can you point out the difference between the composite function $y = \sin(\arcsin x)$ and the function $y = x$?

Reader: Yes, I can. The natural domain of the function $y = x$ is represented by the whole real line. As to the composite function $y = \sin(\arcsin x)$, its natural domain coincides with the natural domain of the function $\arcsin x$, i.e. with $[-1, 1]$. The graph of the function $y = \sin(\arcsin x)$ is shown in Fig. 23.

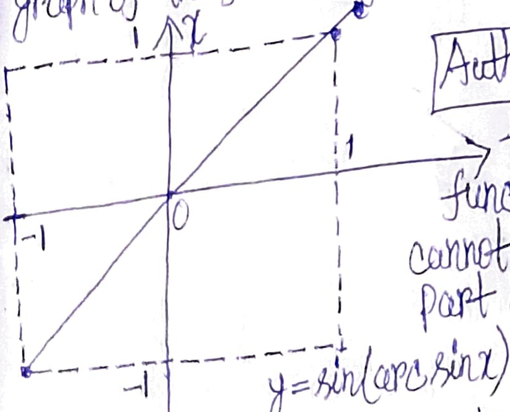

Fig. 23

Author: Very good. In conclusion, let us get back to the problem of the graphical definition of a function. Note that there are functions whose graphs cannot be plotted in principle, the whole curve or a part of it. For example, it is impossible to plot the graph of the function $y = \sin\frac{1}{x}$ in the vicinity of $x=0$ (Fig. 24). It is also impossible to have the graph of the Dirichlet function mentioned above.

Reader: It seemed to me that the Dirichlet function had no graph at all.

Author: No, this is not the case. Apparently, your idea of a graph of a function is always a curve.

Reader: But all the graphs that we have analyzed so far were curves, and rather smooth curves, at that.

Author: In the general case, such an image is not obligatory. But it should be stressed that every function has its graph, this graph being unique.

So the graph of the function (Dirichlet) cannot be constructed but it exists by definition.